

SILVER RESISTANCE IN GRAM-NEGATIVE BACTERIA: INITIAL CHARACTERIZATION OF THE METAL TRANSPORTER SILP

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ABSTRACT

Silver (Ag⁺) is currently one of the most widely used metal to combat microbial infections (1). The antibacterial properties of Ag⁺ are the result of various processes, including its ability to disrupt transcription or to substitute for other metals present in metalloenzymes. However, some Gram-negative bacteria have acquired a resistance mechanism based on active efflux systems that export silver ions from the cytosol. Silver resistance is mediated by the *sil* operon, which encodes nine proteins among which, SilP, a P1B-type ATPase predicted to export Ag⁺ (2). To date, the structure and the function of SilP remain largely unexplored. This project aims to elucidate the molecular mechanism of Ag⁺ transport by SilP. Toward that goal, SilP was expressed in *Escherichia coli* as an eGFP fusion protein, enabling purification by affinity chromatography using anti-GFP nanobodies. Optimized expression and purification conditions yielded sufficient amounts of stable SilP for structural studies. CryoEM analysis produced well-defined 2D class averages and a preliminary 3D reconstruction, demonstrating the suitability of the sample for structural characterization. These results establish a robust purification strategy for SilP and provide the foundation for future high-resolution structural and functional studies of this putative silver transporter.

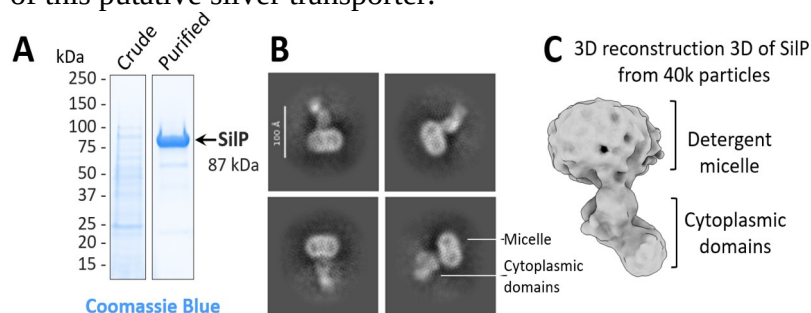


Fig. (A) Coomassie Blue-stained SDS-PAGE analysis of SilP purification. **(B)** Representative 2D classes of SilP in different orientations. **(C)** Preliminary 3D reconstruction of SilP (~40,000 particles)

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